

PALESTINE 6 WNW, TX, USA

WMO: 747490

Lat: 31.7796N

Lon: 95.7232W

Elev: 383

StdP: 14.49

Time Zone: -6.00 (NAC)

Period: 03-19

WBAN: 53968

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	25.5	28.9	11.2	9.9	31.5	15.8	12.4	36.4	13.0	47.3	12.1	46.7	2.9	N/A	0.328

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.5	98.8	73.7	96.6	74.0	94.5	74.3	78.2	89.1	77.4	88.4	76.9	87.7	4.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.4	135.1	81.9	74.8	132.2	81.5	74.0	128.8	81.1	41.8	89.5	41.1	88.8	40.4	87.5	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.7	9.0	8.2	17.9	101.8	4.1	2.8	14.9	103.8	12.6	105.5	10.3	107.1	7.3	109.1
			15.6	79.3	4.6	1.0	12.2	80.0	9.5	80.6	6.9	81.2	3.6	81.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	67.1	48.8	52.5	60.3	66.2	73.5	80.6	83.0	83.9	78.6	68.4	58.1	50.4	(6)
	DBStd	14.59	10.06	10.59	9.15	7.48	6.05	3.75	3.20	3.82	5.08	7.93	9.68	9.80	(7)
	HDD50	416	146	89	24	3	0	0	0	0	0	1	35	118	(8)
	HDD65	1925	510	366	196	75	11	0	0	0	0	58	247	462	(9)
	CDD50	6662	108	159	345	488	729	919	1024	1051	858	572	277	132	(10)
	CDD65	2697	7	15	52	111	276	469	559	586	408	164	39	11	(11)
	CDH74	27211	22	68	265	761	2287	4909	6323	6996	4027	1336	181	36	(12)
	CDH80	12344	0	5	28	141	749	2246	3175	3729	1849	402	19	1	(13)
(14) Wind	WSAvg	3.7	4.1	4.6	4.6	4.3	3.8	3.2	2.9	3.0	3.0	3.3	3.6	3.8	(14)
(15) Precipitation	PrecAvg	46.7	4.1	4.0	3.8	3.7	3.9	4.3	2.7	3.3	3.7	4.5	3.8	4.2	(15)
	PrecMax	65.4	8.2	9.0	7.1	8.7	7.8	9.9	10.8	9.6	7.6	12.4	12.6	9.1	(16)
	PrecMin	26.6	0.4	0.1	0.9	0.7	0.0	0.0	0.0	0.0	1.5	0.0	0.2	0.6	(17)
	PrecStd	10.6	2.2	2.3	1.7	2.0	2.4	2.7	2.3	2.7	1.8	3.4	2.7	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.1	80.3	83.1	88.4	92.7	99.2	100.8	103.6	99.3	90.7	82.8	78.2	(19)
		MCWB	61.7	62.7	64.5	70.6	73.0	74.4	73.8	73.4	71.2	70.4	68.8	65.6	(20)
	2%	DB	72.9	76.0	79.9	83.7	89.5	95.2	97.9	99.7	95.3	87.4	78.9	73.9	(21)
		MCWB	60.4	63.1	64.5	68.7	72.3	74.2	73.9	73.8	72.4	70.6	67.3	65.2	(22)
	5%	DB	69.3	72.2	77.1	81.0	86.8	92.9	95.6	97.2	92.5	84.8	75.6	70.3	(23)
		MCWB	60.3	62.5	63.7	67.3	72.2	74.8	74.6	74.5	72.6	69.2	65.2	63.1	(24)
	10%	DB	65.9	68.8	74.1	78.6	84.4	90.5	93.1	95.0	89.8	81.6	72.7	67.5	(25)
		MCWB	57.7	60.8	62.3	66.0	71.2	75.1	75.3	74.5	71.9	67.6	64.9	60.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.5	67.9	68.9	73.5	76.1	79.0	79.1	79.3	77.6	76.4	71.7	69.4	(27)
		MCDB	72.1	74.5	76.4	81.6	85.7	89.3	89.2	91.1	87.1	82.8	78.5	73.5	(28)
	2%	WB	64.0	66.4	67.6	71.5	75.0	77.9	78.1	78.0	76.5	74.4	69.7	67.1	(29)
		MCDB	69.8	72.4	75.0	79.4	84.8	88.4	88.8	89.9	86.2	81.8	75.4	72.2	(30)
	5%	WB	61.6	64.7	66.4	70.2	74.0	77.1	77.4	77.2	75.5	72.9	68.3	64.8	(31)
		MCDB	66.4	70.2	73.4	78.3	83.6	87.5	88.4	89.0	85.3	80.4	73.6	69.8	(32)
	10%	WB	57.8	62.3	65.0	69.0	73.0	76.3	76.7	76.5	74.8	71.3	66.0	61.6	(33)
		MCDB	63.0	68.0	71.8	76.9	82.1	86.3	87.5	88.0	84.5	78.1	71.9	66.6	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	20.0	18.8	20.0	20.6	18.7	18.2	19.4	20.5	20.4	21.4	20.4	19.8	(35)
		MCDBR	22.7	21.1	20.2	21.0	20.4	21.5	23.1	23.7	23.0	22.2	20.4	21.7	(36)
	5% WB	MCWBR	14.3	12.0	10.6	9.7	7.9	5.8	5.3	5.2	6.7	9.1	11.3	14.3	(37)
		MCWBR	17.5	16.6	17.3	17.7	17.6	17.5	18.6	19.8	19.0	17.6	17.0	18.6	(38)
(40) Clear-Sky Solar Irradiance	taub	taub	0.320	0.341	0.364	0.406	0.441	0.449	0.457	0.459	0.416	0.365	0.354	0.328	(40)
		taud	2.511	2.462	2.407	2.302	2.256	2.273	2.288	2.273	2.345	2.470	2.450	2.498	(41)
	Ebn at Noon	Ebn at Noon	287	291	291	281	271	267	264	263	270	279	272	278	(42)
		Edh at Noon	28	32	36	42	44	43	42	42	38	31	29	27	(43)
(44) All-Sky Solar Radiation	RadAvg	RadAvg	893	1005	1376	1704	1838	2023	2058	1927	1625	1314	960	781	(44)
	RadStd	RadStd	84	177	128	115	138	170	143	136	143	184	119	102	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (40 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
										+203	+158

Nomenclature: See separate page